

ROHIT KUMAR DUBBAKA

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ECE graduate who ships real hardware - from schematic to field deployment. Seeking embedded roles to drive outcomes through system-level knack in the AI world.

EXPERIENCE

DeltaIoT Pvt Ltd <i>Embedded Systems Intern</i>	Feb 2026 – Present <i>Hyderabad</i>
- Designed and validated a compact 4-layer LTE/GPS pluggable module (55×44 mm), replacing the existing 150×170 mm fixed IoT board - 80% board-area reduction with improved RF signal performance during bench validation. - Delivered full PCB design for an IC-based hot-swappable power management breakout board rated for 2.5A current bursts - from schematic capture to production. - Reverse-engineered an industry-grade IoT device pin-by-pin; findings directly informed redesign of power circuitry and selection of superior external ADC modules, producing a reusable reference schematic for the engineering team. - Handled an end-to-end IoT deployment including field installation and firmware updates, gaining exposure to real-world device lifecycle.	

TECHNICAL SKILLS

Languages: Embedded C, Verilog/VHDL
MCUs & Protocols: STM32 (ARM Cortex-M), ESP32 | I²C, SPI, UART, RS-485, USB 2.0, AT Commands
PCB Design: Altium Designer, KiCAD, Differential Pairs, RF Routing
Tools: STM32CubeIDE (HAL), ESP-IDF, Xilinx Vivado, LTspice, DSO/FFT Analysis, Git
Domains: Embedded Firmware, Hardware Bring-Up, Edge AI/TinyML, Power Electronics (Buck/Boost, LDO)

PROJECTS

Aeromesh — CraftifAI Buildathon <i>FirmGen, ESP-IDF, ESP32</i>	<i>Aug 2026</i>
- Rapid-prototyped edge firmware on ESP32 + MPU6050 for in-situ turbulence classification; trained an edge AI model during the buildathon from live motion data and demonstrated real time classification. - Recognized by organizers post-competition — invited for a spotlight interview on build methodology; project featured on CraftifAI's official LinkedIn.	
sEMG-Based Speech-Controlled Wheelchair <i>STM32, Python, LTspice, KiCAD</i>	<i>Jun 2025 – Mar 2026</i>
- Designed a custom analog front-end (AFE) PCB for sEMG acquisition from laryngeal muscles; bench-validated gain, CMRR, and noise floor using DSO/FFT analysis with surface electrodes. - Implemented real-time closed-loop firmware on STM32 integrating ADC, interrupts, and UART; trained a Random Forest classifier for muscle-command recognition. - Delivered full end-to-end system: wearable AFE → signal acquisition → ML inference → H-bridge motor actuation (BTS7960) with safety overrides.	
Priority Arbiter for Multi-Channel Bus <i>VHDL, Basys-3 FPGA</i>	<i>Dec 2024 – Mar 2025</i>
- Designed and validated a deterministic multi-channel bus priority arbiter from RTL to FPGA using encoder and FSM-based sequencer arbitration logic. - Verified timing correctness through RTL simulation and confirmed deterministic behavior under concurrent access on hardware.	

CERTIFICATIONS	Links	EDUCATION	
PCB Design with KiCad – Udemy	Aug 2026	Amrita Vishwa Vidyapeetham	2026
ML Techniques in MATLAB – MathWorks	Jun 2025	B.Tech in ECE	CGPA: 7.6/10
SystemVerilog Fundamentals – Udemy	Dec 2024	VINJEE Junior College, Hyd.	2022
		Class XII – TSBIE	
		HPS Begumpet	2020
		Class X – ICSE	